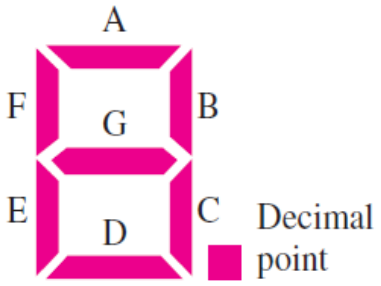
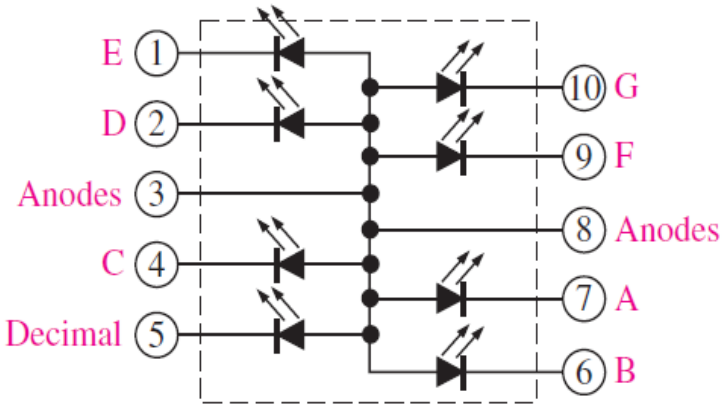
Sheet 5

1. Determine how to connect the seven-segment display in the figure below to display the number [**0**]. The maximum continuous forward current for each LED is 7.9 mA and a +10 V dc source is to be used. The forward voltage-drop across a red LED segment is V_F=2.1 V. Use common anode connection in your design.





2. Develop a yellow LED (V_F=2.5 V for I_F=30mA) traffic-light array using a minimum number of limiting resistors that operates from a 24 V supply and consists of 50 LEDs with and an equal number of LEDs in each parallel branch. Show the circuit and the resistor values.